
Classifying Alzheimer's Disease Progression in Brain MRI Scans Using Convolutional Neural Networks

William Osborne

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Introduction

Alzheimer's Disease is a serious neurological condition that gradually affects memory, thinking and behavior. Identifying Alzheimer's early is crucial because it can help slow down symptoms and improve treatment options. One promising method for early detection involves analyzing brain MRI scans to spot structural differences that appear as the disease progresses. The goal of this project is to explore whether these MRI images can reliably indicate the progression of Alzheimer's from a healthy brain, to various stages of neurodegenerative progression, to Alzheimer's disease itself. This analysis aims to answer the question: "How do structural patterns in MRI scans differ across the four stages of Alzheimer's progression, and which specific imaging features might serve as early indicators of cognitive decline?" The benefit of addressing this question would be the utilization of MRI scans to diagnose Alzheimers earlier by untrained clinicians, helping patients start treatments sooner and potentially slowing the progression of the disease.

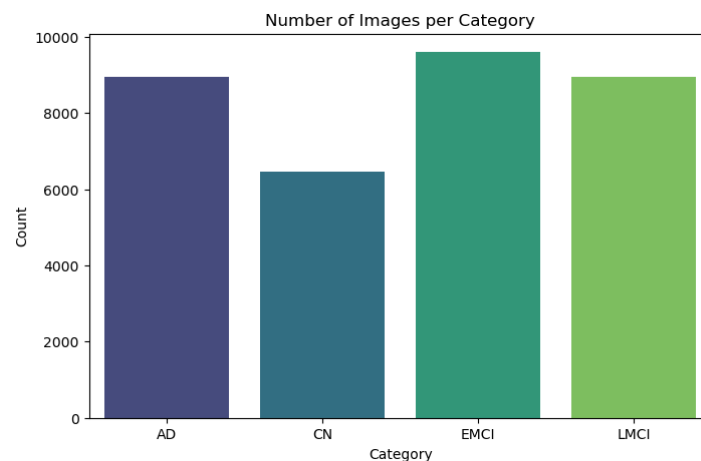
To investigate this problem, an Alzheimer MRI dataset from Kaggle, which includes thousands of MRI scans categorized into four stages of Alzheimer's progression, was used. During the initial exploratory data analysis, key differences in brain structures, like hippocampal atrophy (shrinkage) and cortical thinning, were identified. Existing research supports these differences as early signs of Alzheimers – "The disease [is] most severe in the medial temporal lobe, particularly the entorhinal cortex and hippocampus." (Vemuri & Jack, 2010).

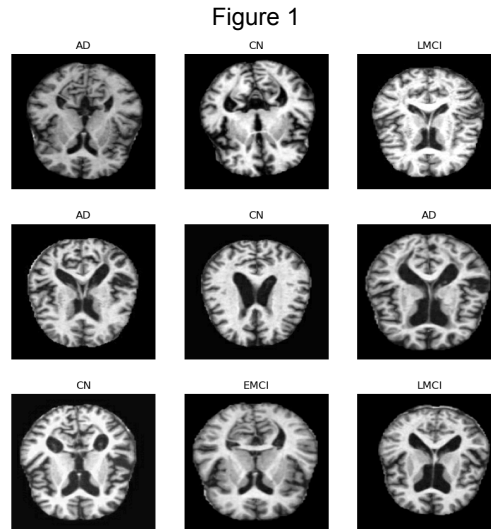
For the classification of MRI brain scans, machine learning models were trained to see how effectively they could classify the MRI scans into the correct Alzheimer's stages. Achieving good accuracy with these models could help doctors and researchers recognize early structural changes that indicate cognitive decline, improving the outcomes for patients.

Methods

Data Source and Description

The dataset primarily used in the project was downloaded from Kaggle, the “ADNI-4C Alzheimer’s MRI Classification Dataset”. This dataset contains MRI scans of the human brain that are categorized into four different groups, based on the progression of Alzheimers in the human brain: CN (Cognitively Normal), EMCI (Early Mild Cognitive Impairment), LMCI (Late Mild Cognitive Impairment), and AD (Alzheimer’s Disease). The dataset is stored as a collection of folders for each target class with thousands of JPEG images (Fig 1), Some images differed slightly in dimensions, but the majority of the images were 190 pixels wide and 200 pixels wide. The images varied slightly in color tint, before being converted to grayscale (Fig 2).





Data Preprocessing

To make sure that all results were consistent and reliable, several preprocessing steps were employed. First, every image was standardized to 190x200 pixels for uniform input dimensions. As mentioned earlier, the images were converted to grayscale, and the pixel values were normalized to a range between 0 and 1. Finally, an augmentation step was added in the beginning of the neural network to artificially vary the image data by applying random rotation, scaling, translation, and contrast factors to each image, for the purpose of preventing overfitting to exactly aligned brain scans, and increased the model's capacity to adapt to new data.

Baseline Analysis / Model

The initial baseline analysis was carried out using a custom Convolution Neural Network (CNN). CNNs were selected based on their abilities to identify visual patterns in image inputs, helping to extrapolate subtle differences between the scans. The baseline neural network written and tweaked included:

- Three convolutional layers with increasing complexity (32, 64 and 128 filters) to capture patterns in each kernel.

- Max pooling layers (2x2) after every convolutional layer to reduce the dimensionality of the feature maps (learned visual details that distinguish between classes).
- Dropout layers (0.25 dropout rate) following each convolutional layer, and a final dropout layer with a 0.5 dropout rate. These layers prevent overfitting the training data.
- A dense output layer with a softmax activation function, calculating the most probable class from the given weights.

Some of the hyperparameters set:

- Optimizer: Adam, defacto for updating model weights using gradient descent.
- Loss function: Categorical cross-entropy for a multiclass dataset.
- Learning rate: Initially 0.01, layer 0.0001 to improve training stability / exploding gradients.
- Gradient Clipping: Clip value of 1.0 to limit magnitude of weight updates during each training pass.
- Epochs: 15

This approach was chosen after research on existing CNNs, combined with efforts to keep loss values stable during training.

Alternative Analysis / Model #1

Another alternative strategy to classify the MRI scans involved transfer learning using existing deep learning models. MobileNetV2 was specifically chosen from tensorflow's library, due to its lightweight model size and its accuracy on a wide range of image classification problems. After researching methods to improve image classifier accuracy, transfer learning was used due to the substantial amount of training and research that had already been invested in existing models, making them potentially faster to fine tune and more effective at generalizing to features they may have seen before. For this approach, the top (dense) layers were stripped from MobileNetV2 and replaced with a dropout layer and a dense layer that were adjusted to

match the number of output classes. Using this model, the modified MobileNet was then fit to the grayscale MRI training samples.

Hyperparameters included:

- Learning Rate: 0.001
- Epochs: 10
- Optimizer: Adam
- Loss Function: Categorical cross-entropy

Alternative Analysis / Model #2

After evaluating the first two models, the decision was made to improve the initial custom CNN by researching how each layer affected the neural network's skill at pattern recognition. One previous study found that "residual learning facilitates the model to capture minor textural and contrast variations [in MRI brain scan images] and also facilitates overcoming the vanishing gradient problem". (Zahoor *et al.*, 2024) With the incorporating residual layers, where the initial input is added to the output after the activation function, the model could retain learned features better across epochs (Fig 3). The ultimate goal of this stage of the analysis was to incorporate more existing research on CNN structure to understand what impacted validation loss and accuracy while training.

Another structural change for the model was the addition of Batch Normalization. One of the most problematic issues encountered was caused by exploding gradients. However, by normalizing the model weights after each Convolution, the training and validation loss were kept much more stable. To help the model find meaningful, general patterns, kernel regularization and weight decay were also incorporated to add a penalty for over-relying on tiny details or pixels. AdamW, a version of Adam that decouples weight decay from the gradient update, was chosen for this task. Finally, global average pooling was inserted at the end of the training pass to prevent overfitting by reducing overall dimensionality.

A final change that improved the scaling of the model includes reducing the filter size range from 32, 64, 128, and 256 to 32, 48, 64, and 96. Evidence has been found that reducing these filter counts would prevent the model from memorizing training data: “[the] introduction of ‘strength’ of filters significantly reduces the number of parameters to learn ... and therefore avoids over-fitting with limited training” (Keshari et al., 2018). The journal indicated that given the problem’s small input shape, large filter sizes would increase overfitting due to the convolutional layer’s ability to memorize specific training data features. This improvement on the custom CNN architecture, by incorporating residual connections and measures to reduce overfitting, aimed to create a more robust and higher-performing model for classification on completely new MRI images.

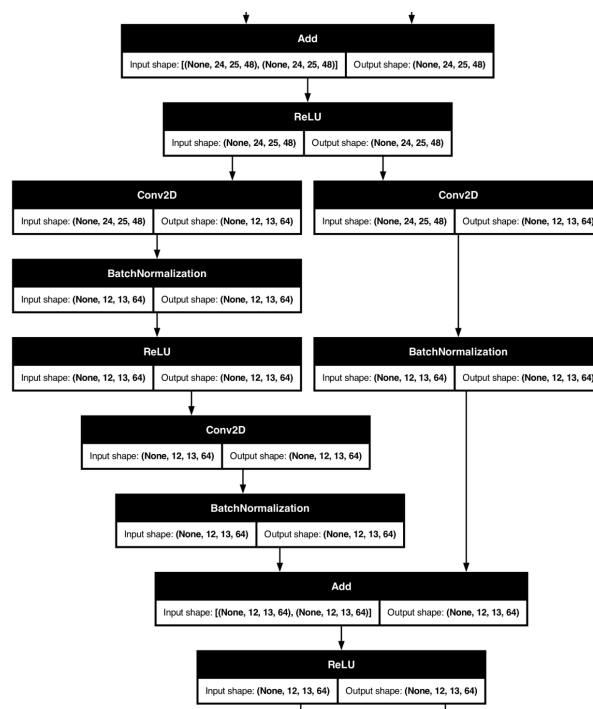


Figure 3: Resnet Block Diagram from Improved CNN

Training and Evaluation

The dataset was split into training and validation sets, with 80% of the data used for training and 20% used for validation. 20% sc Due to resource constraints with training deep neural networks, cross-validation was skipped and a simple train/test split was used for

simplicity and efficiency. Most of the hyperparameters between the two models were manually tuned based on observations during the training process, specifically the learning rates and the dropout rates. The two main evaluation metrics used:

- Validation Accuracy: How well the model could perform on MRI scans it hadn't seen yet.
 $(TP + TN) / \text{Total Predicted}$
- Categorical Cross-Entropy Loss: How well the model's gradients were converging, and how "sure" the model was of a category's probability.

Baseline Model Performance (Custom CNN)

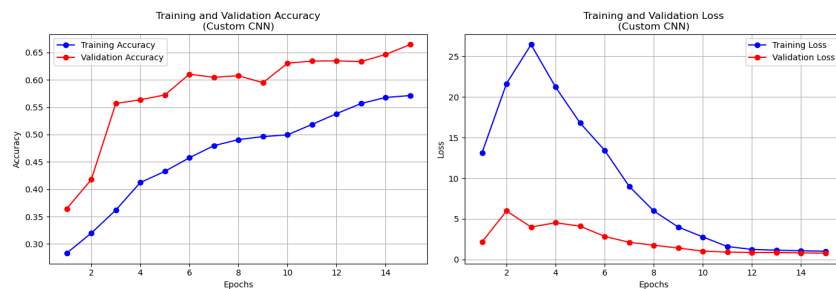


Figure 4: Initial Custom CNN

The initial CNN trained on grayscale MRI images achieved a validation accuracy of around 65% after 15 epochs of training. This is significantly higher than the random guess baseline of 25% for four equally distributed classes, indicating the model was able to identify useful structural features in the MRI scans.

However, the model initially experienced instability during training. Early training runs with a learning rate of 0.01 led to huge fluctuations in loss, suggesting issues with gradient magnitude. Lowering the learning rate to 0.0001 and applying gradient clipping helped stabilize training and reduce loss over time (Fig 4).

Alternative Model Performance (Improved Custom CNN + MobileNetV2)

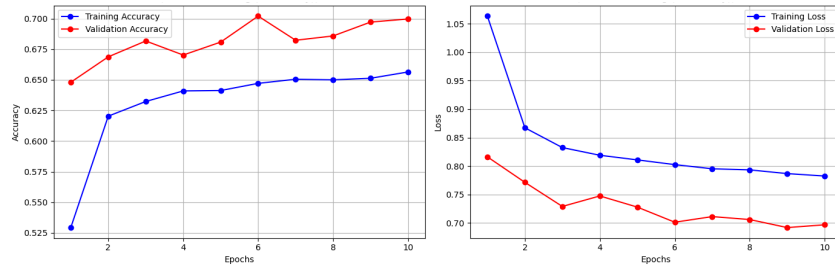


Figure 5: Transfer Learning / Finetune

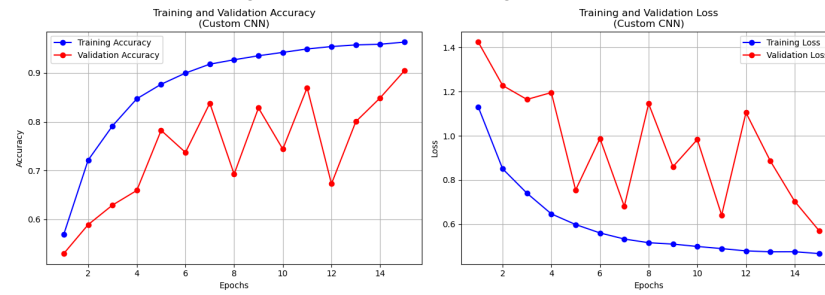


Figure 6: Improved Custom CNN

The later improved custom CNN demonstrated significantly higher performance compared to the initial baseline model. This improved model, incorporating residual layers, scaled-down filter counts, and a more sophisticated optimizer, achieved a validation accuracy of 90% after 15 epochs (Fig 6). This represents a major improvement over the initial model.

The fine-tuned MobileNetV2 achieved a validation accuracy of 70% after 10 epochs (Fig 5). Because this model began with weights pre-trained on a large, diverse dataset (ImageNet), it likely had a better starting point for learning subtle patterns in MRI data. The accuracy and loss curves for the improved custom CNN were more stable compared to the initial custom model, with a clearer positive trendline in accuracy and a stable, decreasing loss. MobileNetV2 also showed more stable training patterns compared to the initial model.

Comparison and Interpretation

Model	Validation Accuracy	Training Epochs	Overfitting	Size (Parameters)	Notes

Custom CNN	65%	15	Mild	50,000	Required tuning of learning rate and dropout
MobileNetV2 Finetuned	70%	10	Minimal	3,400,000	Better initial performance, more stable loss
Improved Custom CNN	90%	15	Minimal	600,000	Incorporated residual layers and weight decay, best performance

These results suggest that the iterative improvements made to the custom CNN architecture were highly effective. The incorporation of residual layers helped the model learn deeper features and improve gradient flow during training. The adjusted filter counts and the use of AdamW with weight decay contributed to better regularization, reducing overfitting and improving generalization performance.

The improved custom CNN significantly outperformed both the initial custom CNN and the fine-tuned MobileNetV2 model in classifying Alzheimers stages. Overall results proved that

a domain-specific neural net specifically optimized for the input type and task can be a lot more effective than relying on existing models adapted to a task with transfer learning.

Discussion and Conclusion

This research project aimed to explore the potential of CNN's for real-world classification tasks that would normally require a trained physician to perform. During the process of training and evaluating several models on the ADNI-4C dataset, it was an aim to explore if sophisticated neural networks could pick up on small patterns that the untrained human eye cannot. The initial baseline analysis, using a custom convolutional network, provided a good understanding of how a simple, generic CNN could handle basic feature recognition and discern between cognitive stages. With a validation accuracy of 65%, significantly higher than random chance, the model proved that structural information in the provided MRI scans does contain these discernible patterns. This stage laid the groundwork for improving on its limited capabilities, even though it's accuracy is well below what is required for a real-world medical tool. It also highlighted more significant problems to solve, such as the training instability and overfitting.

The alternative analysis, starting with a transfer-learning adapted MobileNetV2, was a noticeable improvement over the simple CNN. With a higher validation accuracy and a more stable training curve, it highlighted the benefits of using established features trained on diverse datasets. However, it also led me to believe that a domain-specific neural net with a more sophisticated architecture could still outperform both, prompting me to research more of the different types of layers and established model research in the medical imaging field. Other scientists have also reported success in this approach – “In this work our main objective was to prove that a simple CNN could achieve top performance if its design and training is domain-driven .. [and] achieved top performance in many evaluation criteria” (Simantiris & Tziritas, 2020).

The final alternative analysis model, the overhauled custom CNN, proved to be the most effective model tested, achieving a validation accuracy of 90%. The enhancements, including residual connection, kernel regulations, scaled filter counts, and weight decay addressed a lot of the prior limitations of the previous two models, using proven techniques in this space.

“Residual neural networks have demonstrated robust potential for clinical diagnosis across diseases like lung cancer, melanoma, and Alzheimer’s, making them strong candidates for real-world diagnostic tools” (Xu, Fu, & Zhu, 2023). These changes led to significantly more stable training and higher generalization performance on unseen data. Using existing research to tailor the model to the task was rewarded with much better results, and proving viability for diagnosing real-world patients with a high degree of accuracy.

Despite promising results from the analysis, there are limitations to practicality and further development. The dataset contains images only from a single source, and contains only grayscale colors, while MRI imaging contains numerous channels from different signal intensities. The relatively small size has the potential to be expanded with data from other sources, as well as different scanner types and a broader patient demographic. Computational limits on my current machine limited the ability to use k-fold cross-validation, which may have enhanced the training process. And despite achieving high accuracy when distinguishing features, the process of understanding how and why it classified scans is difficult to understand in a complex model such as this, making it difficult for physicians to interpret its findings.

In conclusion, this project successfully demonstrated that structural patterns in brain MRI scans are distinct across the progression stages of Alzheimer's disease, from cognitively normal to full AD. Consequently, it showed advanced machine-learning techniques could use these distinct structural patterns to predict disease progression in a statistically effective manner. The high performance of the improved CNN shows promise as an applied medical tool, not just a research project. Continuing to refine and study these models across more domain-specific

tasks, can help researchers move closer to tools that can outperform human doctors, allowing for earlier diagnoses and increasing quality of life for patients with this challenging disease.

Works Cited

- Keshari, R., Vatsa, M., Singh, R., & Noore, A. (2018). Learning structure and strength of CNN filters for small sample size training. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 9349–9358).
- Simantiris, G., & Tziritas, G. (2020). Cardiac MRI segmentation with a dilated CNN incorporating domain-specific constraints. *IEEE Journal of Selected Topics in Signal Processing*, 14(6), 1235–1243. <https://doi.org/10.1109/JSTSP.2020.3013351>
- Vemuri, P., & Jack, C. R. (2010). Role of structural MRI in Alzheimer's disease. *Alzheimer's Research & Therapy*, 2(1), 23. <https://doi.org/10.1186/alzrt47>
- Xu, W., Fu, Y.-L., & Zhu, D. (2023). ResNet and its application to medical image processing: Research progress and challenges. *Computer Methods and Programs in Biomedicine*, 240, 107660. <https://doi.org/10.1016/j.cmpb.2023.107660>
- Zahoor, M. M., Khan, S. H., Alahmadi, T. J., Alsahfi, T., Mazroa, A. S. A., Sakr, H. A., Alqahtani, S., Albanyan, A., & Alshemaimri, B. K. (2024). Brain tumor MRI classification using a

novel deep residual and regional CNN. *Biomedicines*, 12(7), 1395.

<https://doi.org/10.3390/biomedicines12071395>